

Solutions to Quiz 8

Question 1

Consider the polynomial $f(X) = X^3 - 2X^2 + 12X - 6$ as a polynomial in the ring $\mathbb{Z}_p$ of p -adic integers for the following values of p .

- a. $p = 2$
- b. $p = 5$
- c. $p = 3$

In each case describe the structure of the decomposition of $\mathbb{Z}_p[X]/\langle f(X) \rangle$ into domains.

(**Hint:** Observe that $f(1)$ is divisible by 5 and $f(2)$ is divisible by 9.)

Solution:

- a. We note that $f(X)$ is irreducible in $\mathbb{Z}_2$ by Eisenstein's criterion. Moreover, $f(X) \bmod 2 = X^3$. It follows that $\mathbb{Z}_2[X]/\langle f(X) \rangle$ is a ramified extension of degree 3 of $\mathbb{Z}_2$. In other words, it contains a prime π such that $2 = u \cdot \pi^3$ where u is a unit.
- b. Since $f(1)$ is divisible by 5 we can divide $f(X)$ by $X - 1$ in $\mathbb{F}_5$ to obtain the identity

$$f(X) \bmod 5 = X^3 - 2X^2 + 2X - 1 = (X - 1)(X^2 - X + 1)$$

Since $X^2 - X + 1$ has no roots modulo 5, it is irreducible. This shows that

$$\frac{\mathbb{F}_5[X]}{\langle f(X) \rangle} \cong \frac{\mathbb{F}_5[X]}{\langle (X - 1) \rangle} \times \frac{\mathbb{F}_5[X]}{\langle X^2 - X + 1 \rangle} \cong \mathbb{F}_5 \times \mathbb{F}_{5^2}$$

By Hensel's lemma, we see that $f(X) = (X - \alpha)g(X)$ for some $\alpha \in 1 + 5\mathbb{Z}_5$ and $g(X) \in \mathbb{Z}_5[X]$ an irreducible quadratic polynomial lifting $X^2 - X + 1$ from $\mathbb{F}_5[X]$. This corresponds to the decomposition:

$$\frac{\mathbb{Z}_5[X]}{\langle f(X) \rangle} \cong \mathbb{Z}_5 \times \frac{\mathbb{Z}_5[X]}{\langle g(X) \rangle}$$

where 5 is a prime in the last factor.

- c. Since $f(2)$ is divisible by 9 we divide $f(X)$ by $X - 2$ to obtain

$$f(X) \bmod 9 = (X - 2)(X^2 + 12)$$

Since 2 is not a root of $X^2 + 12$ modulo 9, we see that there is a β in $\mathbb{Z}_3$ which lifts $2 \bmod 9$ and is a root of $f(X)$. Thus we can write $f(X) = (X - \beta)h(X)$ where $h(X) \bmod 9 = X^2 + 12$.

We see that $X^2 + 12$ has no roots modulo 9 (since a root must be of the form $3a$ and no such number is a root). It follows that $h(X)$ is irreducible in $\mathbb{Z}_3$. (We can also deduce this from Eisenstein's criterion.).

Thus we have the decomposition:

$$\frac{\mathbb{Z}_3[X]}{\langle f(X) \rangle} \cong \mathbb{Z}_3 \times \frac{\mathbb{Z}_3[X]}{\langle h(X) \rangle}$$

where the last factor is a ramified extension of degree 2 of $\mathbb{Z}_3$. In other words, this factor contains a prime η such that $3 = w \cdot \eta^2$ and w is a unit.